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Ecofriendly Degradation of PET via Neutral Hydrolysis: Degradation Mechanism and Green Chemistry Metrics

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Abstract: Waste polyethylene terephthalate (PET) bottles represent 12% of global plastic waste; however, only 9% are recycled. Hydrothermal processing presents the opportunity to upcycle waste PET into its monomers, particularly, terephthalic acid (TPA). In this study, post-consumer PET sparkling water bottles were neutrally hydrolysed via a hydrothermal process operating within a temperature range of 220–270 °C, a residence time of 30–90 min, and autogenous pressure of 25–90 bar. Under these conditions, the TPA yield varied between 7.34 and 81.05%, and the maximum TPA yield was obtained at 250 °C, 90 min, and 40 bar. The process temperature had a more profound impact on the PET conversion and TPA yield than the residence time. The values of the environmental factor (EF) were found to be 0.017–0.106, which were comparable to those of bulk chemicals (EF < 1). With the chosen operating conditions, the environmental energy impact (EEI) of TPA production was estimated to be 5.29×10^4 °C min. The findings demonstrate that neutral hydrolysis is a feasible approach for converting PET polymers into monomers under mild environmental conditions. In addition, a GCMS analysis of the aqueous-phase product revealed a notable increase in the secondary degradation products of TPA, such as benzoic acid, rising from 66.4% to 75.7% as the process temperature increased from 220 °C to 270 °C. The degradation mechanisms of PET were found to be decarboxylation, dehydration, and oxidation. The dominant mechanism was found to be a decarboxylation reaction.

Keywords: terephthalic acid; chemical recycling; environmental factor; environmental energy impact; decarboxylation



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1. Introduction

In modern-day life, plastic-based materials have become an unavoidable commodity in every aspect of human life. A recent study by Vuppaladadiyam et al. mentioned that global plastic production reached ca. 367 Million Metric Tonnes (MMT) in 2020, and is expected to grow at an annual compound rate of 3% and reach 600 MMT by 2050 [1,2]. However, only 9% of total plastic waste is recycled; 12% is incinerated and 79% accumulates in landfills.

The rise in the production of plastic waste and its disposal in landfills represents a global environmental challenge. Many nations worldwide are struggling to combat an increase in the volumes of plastic waste being discarded and the omnipresent plastic

pollution. It is worth noting that an increase in the demand for plastic correlates with higher fossil fuel consumption, energy requirements, and greenhouse gas emissions [1]. Additionally, plastics possess unique characteristics, including durability, stability, multifunctionality, and are lightweight, making them excellent alternatives to steel, paper, etc. The most common forms of fossil-derived plastics include polyethylene (PE), polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), polyurethane (PU), and polyvinylchloride (PVC) [2].

PET is a commonly used polymer, with an annual production of more than 70 MMT [3]. PET is extensively used in manufacturing products, such as water bottles, textiles, and food packaging [4]. Mechanical, chemical, and biochemical recycling are generally considered for recycling PET, with mechanical recycling being the most industrially established. However, after multiple recyclings, the quality of the recycled PET material may be inferior, making mechanical recycling challenging. Therefore, they are downcycled into inferior products, such as carpets, which eventually end up in landfills at the end of their life [5]. Unlike mechanical recycling, chemical recycling can be a sustainable route that converts PET polymers into monomers—terephthalic acid (TPA) and ethylene glycol (EG)—or other value-added products. The commonly used chemical recycling methods include neutral hydrolysis [6–8], alkaline hydrolysis [9,10], acidic hydrolysis [11,12], methanolysis [13,14], aminolysis [15,16], and hydrogenolysis [17,18].

Alkaline hydrolysis uses concentrated alkali reagents, such as sodium hydroxide and potassium hydroxide, for PET hydrolysis under a long residence time (hours to days) [19]. The reaction usually leads to the formation of sodium/potassium terephthalate and EG. Subsequently, an acid treatment of the sodium/potassium terephthalate stream is needed to obtain high-purity TPA. In the case of acid hydrolysis, concentrated mineral acids, such as sulfuric acid or nitric acid, are used to produce TPA directly, minimising the need for extensive downstream treatment [20]. However, the primary concerns related to acid hydrolysis include contamination risk, equipment corrosion, the formation of detrimental secondary wastes, and inorganic salts. In addition, the carbonation of EG and dehydration, leading to a lower TPA yield, are the other demerits of acidic hydrolysis [8]. On the other hand, neutral hydrolysis occurs without the use of a concentrated acid/alkali reagent, thereby reducing the concerns related to equipment corrosion, the formation of inorganic salt byproducts, and the reactivity of TPA with the acid or base. Therefore, neutral hydrolysis is generally preferred in comparison with other hydrolysis approaches, namely acid hydrolysis, alkaline hydrolysis, methanolysis, and aminolysis [21,22]. The process conditions employed in neutral hydrolysis include temperatures between 220 and 300 °C, a residence time between 30 and 90 min, and an autogenous pressure between 25 and 90 bar. The water-to-PET ratio during neutral hydrolysis processes has been consistently taken as 10:1 (*w/w*) in the previous literature. This is because 1 mole of PET always requires 2 moles of water to be converted into terephthalic acid and ethylene glycol based on the stoichiometric mole balance. But, according to the previous literature, it has been suggested that a 10-to-1 (water-to-PET) mass ratio is the most effective ratio for the near complete conversion of PET to TPA under neutral hydrolysis conditions without the use of catalysts [23–25]. In addition, neutral hydrolysis can be carried out using hot steam or water and in the presence of the water-soluble salts (mono-hydroxy-ethyl terephthalate and bis-hydroxy-ethyl terephthalate) generated during the hydrolysis of PET. Under neutral hydrolysis conditions, PET is degraded in two phases: (i) a solid phase that is rich in TPA and (ii) an aqueous phase that contains ethylene glycol (EG) and its derivatives. The post-treatment of the solid-phase product involves an alkaline treatment followed by an acid treatment to recover high-purity TPA [26].

Previous articles have focused on the effects of the processing conditions, such as temperature [7,27], residence time [9,28], PET-to-water ratio [29,30], stirring speed [19,31], and autogenous pressure [32], on the hydrolysis of PET into TPA. Other studies have concentrated on the chemical composition of the aqueous phase, aimed at understanding the PET degradation mechanism and the formation of secondary byproducts [24,33]. In addition, the literature published to date has reported either the PET conversion or TPA yield, but not both [11,12,31]. However, it is important to note that the PET conversion does not necessarily equate to the TPA yield, particularly under neutral hydrolysis conditions, where post-treatment of the solid phase is required to recover high-purity TPA. PET conversion is often accomplished through a neutral hydrolysis process and reflects the fraction of the solid product generated from the hydrolysis of the given PET feed mass. It is a simplistic assumption that all the PET converted during the neutral hydrolysis process is TPA without considering the efficiency of the downstream TPA recovery process. Typically, the PET conversion is generally higher than the TPA yield, so reporting both data is crucial to understanding the performance of the conversion and recovery processes and for fair comparison with other hydrolysis techniques.

The current work focused on the neutral hydrolysis of PET, and we investigated the effect of temperature and residence time on both the PET conversion and TPA yield. The TPA derived from a neutral hydrolysis process following an acid and alkali post-treatment was comprehensively characterised using XPS, FTIR, and NMR analyses to assess the carbon and oxygen chemistry, functional groups, and purity, respectively, in comparison to a commercial TPA product. The environmental energy impact, a green chemistry metric that evaluates the effect of PET degradation via neutral hydrolysis on the environment, was determined. Green chemistry metrics were used to determine the economic and environmental viability of the synthesis process. Lastly, the aqueous-phase product was characterised by a GCMS analysis to determine the composition of and provide insights into the PET degradation byproducts formed during the hydrothermal process.

2. Materials and Methods

2.1. Materials

PET bottles (600 mL) were purchased from the canteen at RMIT University, Bundoora campus, Victoria, Australia. The labels and caps were detached from the PET bottles, after which they were manually cut into small quadrilateral fragments, with average dimensions of 10 mm × 10 mm, using scissors. Milli-Q water was used as the solvent for the neutral hydrolysis. Sodium hydroxide (ACS Reagent, ≥97.0%), sulphuric acid (ACS reagent, 95.0–98.0%), and terephthalic acid (98%) were purchased from Sigma Aldrich Pty Ltd, Melbourne, Victoria, Australia. Undenatured ethyl alcohol (100%) was purchased from Chem Supply, Adelaide, South Australia, Australia. Dimethyl sulphoxide—D₆ (DMSO, 99.9%) was purchased from Novachem, Melbourne, Victoria, Australia. High-pressure stirred tank autoclave reactor (SS-316L) was purchased from InstruFlow Pty Ltd, Melbourne, Victoria, Australia designed to withstand a temperature of 500 °C, pressure of 200 bar, and a volume of 500 mL, was used for all the experiments.

2.2. Methods

2.2.1. Neutral Hydrolysis

The hydrothermal reactor was fed with 20 g of PET flakes and 200 mL of Milli-Q water. The hydrolysis reaction was conducted between 220 and 270 °C with a 30–90 min residence time. The autogenous pressure during the reaction ranged from 25 to 90 bar. The reactor was purged with a high-purity N₂ stream to create an inert atmosphere.

The reactor and its contents were heated to the desired temperature (220, 250, 270, or 300 °C) at a constant heating rate of 7 °C/min. Once the reactor reached the target temperature, the reaction was continued for a specified residence time (30, 60, or 90 min) under continuous stirring at 100 rpm. At the end of the residence time, the reactor was cooled rapidly to 25 °C with the help of a chiller. The product was transferred from the reactor into a beaker and vacuum-filtered to separate it into solid and aqueous phases. The solid phase was treated with 2 M NaOH at 150 °C for 20 min, followed by vacuum filtration to separate the solid residue (partially converted PET) and liquid stream (containing Na₂-TPA). The solid residue was collected in a silica crucible and dried overnight. The Na₂-TPA-rich stream was titrated with 1 M H₂SO₄ until a white residue, referred to as TPA, was formed, which was later dried at 105 °C for 24 h. Figure 1 summarises the standard procedure employed to recover TPA and aqueous phase derived from the hydrolysis of PET, along with their respective characterisations. The PET conversion and TPA yield were estimated using Equations (1) and (2) [25].

$$\text{PET conversion (\%)} = \left(1 - \frac{\text{weight of Partially converted PET}}{\text{Initial weight of PET}} \right) \times 100 \quad (1)$$

$$\text{TPA yield (\%)} = \frac{(\text{Weight of TPA})}{(0.8653 \times \text{Initial weight of PET})} \times 100 \quad (2)$$

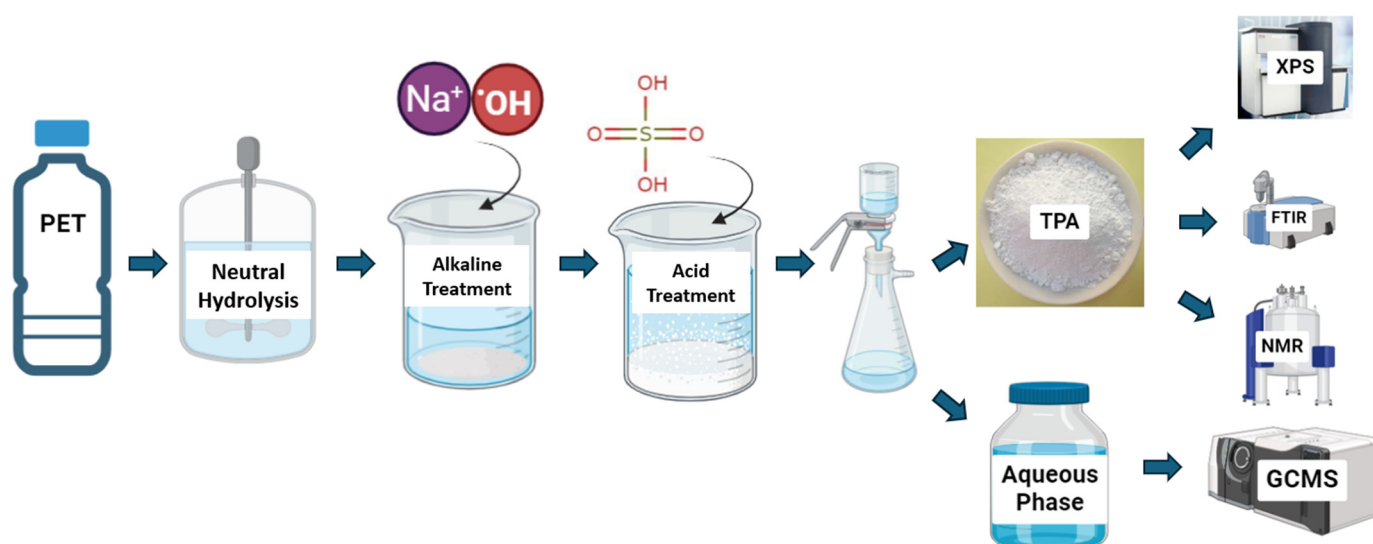


Figure 1. Procedure for product separation and analysis following neutral hydrolysis of PET.

2.2.2. Green Chemistry Metrics

During the hydrolysis of PET, several metrics can be used to assess the economic and environmental impact of the process. The integral factors are the environmental factor (EF), economy energy coefficient (EEC), and a combination of both, which is the environmental energy impact (EEI). EF is a dimensionless factor used in green chemistry and corresponds to the quantity of waste used per unit of desired product [34].

The mathematical expression for calculating the EF is provided in Equation (3). The modified form of Equation (3) indicates that 10% of the mass of the solvent (water) needs to be substituted as fresh feed for subsequent runs [35]. The catalyst and other substitutes

were not considered for evaluation since they were not used during the hydrothermal treatment. The modified form of the EF is represented by Equation (4) [3,29].

$$EF = \frac{(\text{Quantity of waste used})}{(\text{Quantity of desired product})} \quad (3)$$

$$EF = \frac{\left[0.1 \times \left(\frac{\text{Solvent}}{\text{PET}} \text{ ratio}\right)\right] \times \text{Mass of PET}}{\frac{\text{Yield of TPA}}{\text{Maximum theoretical Yield of TPA}} \times \frac{\text{Molar mass of TPA}}{\text{Molar mass of PET}} \times \text{Mass of PET}} \quad (4)$$

EEC ($^{\circ}\text{C}^{-1} \text{ min}^{-1}$) is the energy required to maintain the process conditions (temperature and residence time) to produce the maximum desired product (TPA). The EEC value denotes the effectiveness of the degradation technique being used and is given by Equation (5) [36,37].

$$EEC = \frac{\text{Quantity of TPA}}{\text{Temperature} \times \text{Time}} \quad (5)$$

EEL ($^{\circ}\text{C min}$) is a metric that evaluates the environmental and economic impacts associated with PET hydrolysis and is calculated using Equation (6) [36]. An increase in the TPA yield under less severe conditions will increase the EEC, leading to a lower EEL. The best approach will have a lower EF and EEL, and higher EEC.

$$EEL = \frac{EF}{EEC} \quad (6)$$

2.3. Products Characterisation

X-ray photoelectron spectroscopy (XPS) analysis was carried out on a thermoscientific K- α XPS with monochromatic aluminium under a photon energy of 1486.7 eV and spot size ranging between 30 and 400 μm . The K- α was equipped with a flood gun to flood the analysis chamber with low-energy electrons and Ar⁺ ions for charge neutralisation. Additionally, the equipment had an etching gun that could be operated at 200–3000 eV with three current modes (low, medium, and high). The peaks for carbon and oxygen were referenced at 284.73 eV and 531.98 eV. The instrument analysed the surface of the material within 1–10 nm. The carbon and oxygen fractions were quantified using Casa XPS version 2.1.0.1. Fourier transform infrared spectroscopy (FTIR) was performed using a Thermo Fisher Scientific Nicolet iS20 FTIR spectrometer. The FTIR spectra were captured in absorbance mode over a wavelength of 4000 to 400 cm^{-1} , with the maximum uncertainty of the instrument being 2 cm^{-1} to assess the chemical functional group in the TPA derived from PET. Bruker Ultra Shield 300 Magnet Bruker Advance III Console was used to perform nuclear magnetic resonance (NMR) spectroscopy. The solid TPA derived from PET neutral hydrolysis was dissolved in DMSO as part of the standard sample preparation procedure for NMR analysis. The dissolved solution was transferred into sample vials for NMR analysis. The NMR was used to determine the purity of the derived TPA from PET in comparison to commercial TPA sample. Gas chromatography–mass spectrometry (GCMS, Agilent 8860/5977, Beijing, China) instrument was used for performing the GCMS analysis. Before performing the analysis, the sample was prepared by placing 10 mL of the aqueous-phase sample in an oven at 105 $^{\circ}\text{C}$ for 24 h to remove water. The obtained solid residue was dissolved in ethanol. Then, the dissolved samples were injected into the capillary column and helium was utilised as the carrier gas. During the GCMS analysis, to separate the compounds, the instrument had an HP-5MS (19091S-433UI) capillary column (LxI.D.xt—30 m $\times$ 250 μm $\times$ 0.25 μm). The oven temperature program was as follows: (i) isothermal hold at 78 $^{\circ}\text{C}$ for 3 min, (ii) ramp-up at 10 $^{\circ}\text{C}/\text{min}$ to 250 $^{\circ}\text{C}$, and (iii) isothermal hold at 250 $^{\circ}\text{C}$ for 5 min. Mass spectrometry was used to detect the chemical components that were eluted from the column following spectra matching to NIST library.

The compounds' concentrations were taken as the GC peak area (%) after normalisation of all identified compounds' peak areas. After the GCMS analysis, the identified chemical compounds were used to provide insight into PET's degradation pathways and to identify the primary and secondary degradation products under the given process conditions (supplementary material S1).

3. Results

3.1. Effect of Temperature

The PET hydrolysis was performed at different temperatures with the same residence time. The PET conversion substantially increased from 63 wt.% to 55 wt.% when the temperature was increased from 220 to 270 °C, at a constant residence time of 30 min. As a result of increasing the temperature beyond the melting point of PET (235–250 °C), a higher degradation leading to lower PET conversion was noticed due to an increase in the residual weight of the PET [38]. At 60 min residence time, an increase in the temperature from 220 to 250 °C increased the TPA yield from 12.55% to 45.43%, as shown in Figure 2. The increase in the TPA yield may have been due to the decline in the dielectric constant of water, making water behave as a non-polar solvent enhancing the hydrolysis of PET to TPA [39]. Another plausible reason could be the impact of the reaction temperature on the rate constant when close to the melting point of PET ($T_m = 250$ °C), thus enhancing the mixing of the macromolecules and water [40]. A further increase in the reaction temperature from 250 to 270 °C at a constant residence time of 60 min only slightly enhanced the TPA yield to 49.12% from 45.43%. As the pressure increased from 40 bar to 62.5 bar, there was no substantial effect on the dielectric constant of water, which could be the reason for the increase of 3.69 wt% [39]. It is noteworthy that the dielectric constant of water is higher at pressures above 300 bar.

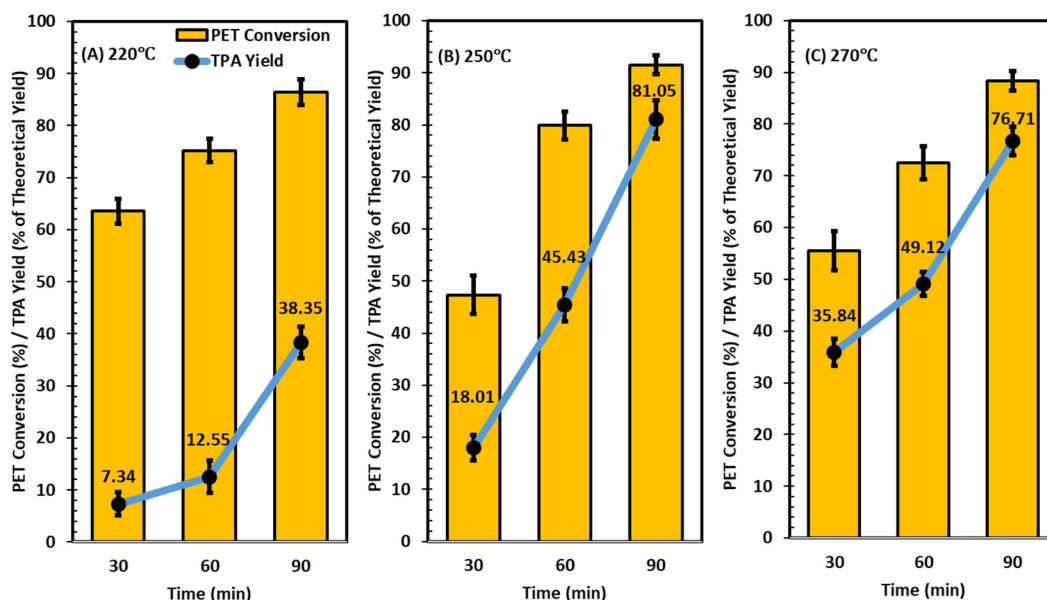


Figure 2. Effect of temperature (220–270 °C) and time (30–90 min) on PET conversion and TPA yield.

Similar results were observed when the process was run in a temperature range of 220–250 °C, maintaining a residence time of 90 min, where the TPA yield increased from 38.35% to 81.05%. However, the TPA yield declined from 81.05% to 76.71% when the temperature was increased from 250 to 270 °C at a residence time of 90 min. Although the increase in the temperature from 250 to 270 °C increased the pressure from 40 to 62.5 bar, the following reasons may be considered responsible for the decline in the TPA yield:

(i) TPA is an unstable compound and may have degraded into a more stable compound after being decarboxylated to form benzoic acid [33], (ii) the molecular rearrangement of the TPA (1,4-benzene dicarboxylic acid) may have led to the formation of isophthalic acid (1,3-benzene dicarboxylic acid), and (iii) the derived TPA from the neutral hydrolysis could have reacted with the generated ethylene glycol to form BHET. Additionally, the decarboxylation of benzoic acid to form benzene or phenols could be another reason for the reduction in the TPA yield at 270 °C compared to 250 °C at 90 min residence time. This is because the produced phenol might have reacted with the TPA derived from the PET, leading to the formation of a methyl enol bridge or resole. Therefore, the optimised temperature was noted as 250 °C.

3.2. Effect of Residence Time

By maintaining the optimised temperature, the optimised residence time for PET hydrolysis was identified by varying the residence times of the PET hydrolysis. At 250 °C, when the residence time was increased from 30 min to 90 min, the PET conversion was noticed to increase from 47.33 wt.% to 91.55 wt.%. The upper-bound results in this study are comparable with an 87% PET conversion reported at 250 °C under neutral hydrolysis conditions [41]. The higher PET conversion may be due to the availability of extra time for the electrically charged radicals, such as the OH^- ions, to act as active intermediates for the decomposition of the PET [42]. At 250 °C, the increase in residence time from 30 to 90 min increased the TPA yield from 18.01% to 81.05%, as shown in Figure 2, corresponding to the results reported in the literature [24,33]. A plausible reason for the rise in the TPA yield can be explained based on the collision theory. A higher number of molecular collisions is likely to happen at 90 min, leading to the formation of more nucleophilic functional groups attacking the carbonyl ester of the PET, leading to a split in the ester group. This split in the polymer molecules happens until the PET is completely degraded into TPA and EG at the optimal degradation temperature of 250 °C [43]. Similar results were observed for the PET hydrolysis at 270 °C and a residence time between 30 and 90 min, where the TPA yields increased from 35.84% to 76.71%. Therefore, the optimised residence time was observed to be 90 min, based on the observed trend in the PET conversion and TPA yield. It can be observed that both the PET conversion and TPA yield were dependent on the residence time.

3.3. XPS Analysis of TPA Derived from PET Hydrolysis

3.3.1. Carbon Chemistry

The measured carbon is 65.56% for a binding energy of 284.73 eV, as shown in Figure 3, which is comparable to the theoretical calculation of carbon in PET, which is 71.40% [38]. The peak at 288.98 eV corresponds to the $\text{C}=\text{O}$ peak, which is carboxylic acid, at 16.57%. The presence of 5.67% $\text{C}-\text{O}$ in the TPA is due to additives, which include antifreeze products (limonene, 2-butanone, and acetic acid), flame retardants (organophosphorus compounds) [44], ultraviolet absorbers (UVA 3034 and UVA3026), and others (acetone, 1,3-dioxolane, and butanal) [45,46]. The values of the carbon peaks are identical to those of the commercial TPA.

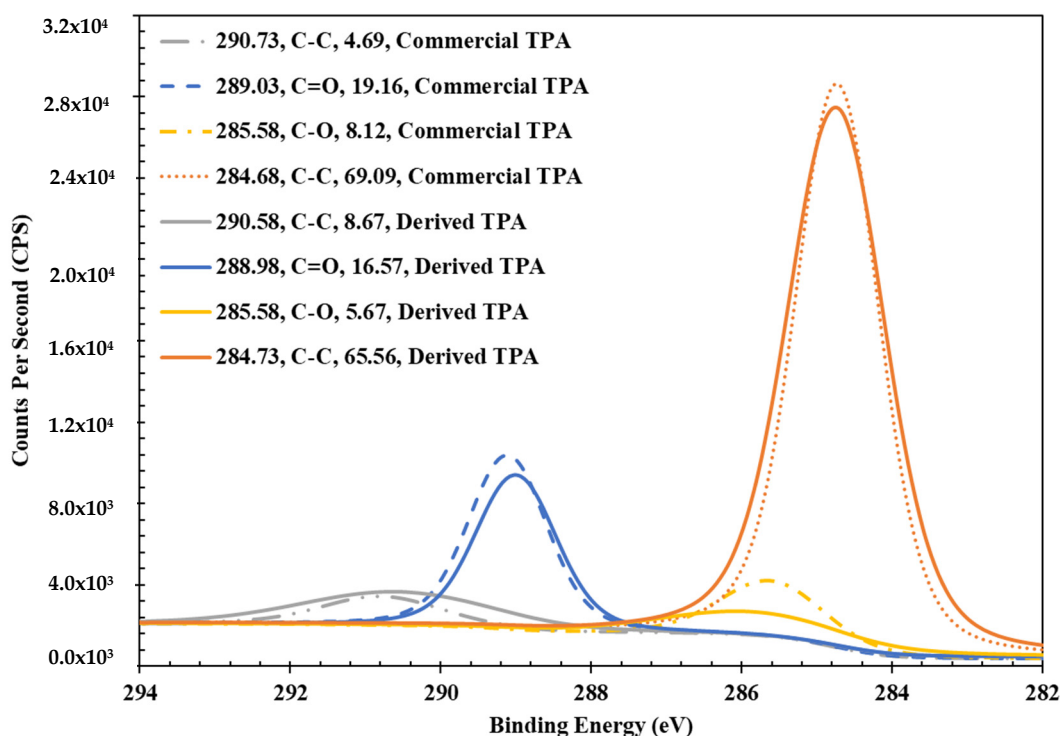


Figure 3. Carbon chemistry of commercial TPA vs TPA derived from neutral hydrolysis.

3.3.2. Oxygen Chemistry

The peak at 533.08 eV is 36.36%, as shown in Figure 4, corresponding to the oxygen attached to the alcohol group (-OH). Another peak at 531.78 eV is 50.12%, attributed to the carboxylic acid (-COOH) in the TPA [23]. The spectra values of the material obtained from the neutral hydrolysis process closely mirror those of commercially available TPA, indicating that the PET-derived product is TPA.

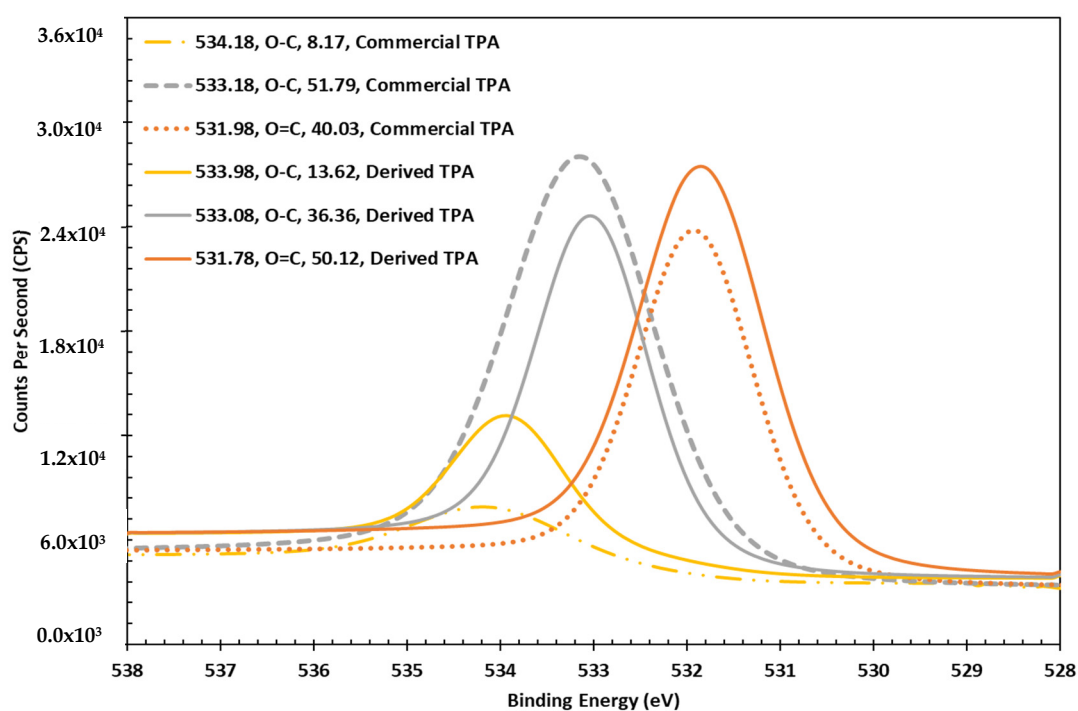


Figure 4. Oxygen chemistry of commercial TPA vs TPA derived from neutral hydrolysis.

3.4. FTIR Analysis of TPA Derived from PET Hydrolysis

The structure of the TPA derived from the neutral hydrolysis of PET was characterised by an FTIR analysis, as shown in Figure 5. The absorbance peak between 2520 and 3110 cm^{-1} can be attributed to hydroxyl stretching [47]. The strong absorbance peaks at 1680 cm^{-1} and 1280 cm^{-1} correspond to the stretching vibration of the C=O in the carboxyl group and the vibration of C-O [32]. Furthermore, the peaks are also strong at 725 and at 878–923 cm^{-1} and can be ascribed to the para-substituted benzene ring of TPA, and the para-substituted arene group of TPA, respectively [48]. On the contrary, weak absorbance peaks are noticeable at 1580, 1510, and 1420 cm^{-1} , corresponding to the vibration of the aromatic ring skeleton in the TPA. In addition to these, the weak absorbance peaks at 1110, 1020, and 447–523 cm^{-1} correspond to the 1,4-substituted benzene rings [26]. The results prove that the TPA derived from neutral hydrolysis show a similar trend to that of commercial TPA. The raw PET absorbance peaks at 1710 and 1250 cm^{-1} are analogous to the stretching vibrations of the C=O and C-O of the ester bonds, respectively [28]. None of these peaks are observed for the partially converted PET (solid phase) formed after the alkaline treatment of the solid phase derived from the hydrolysis of PET. This confirms that the raw PET was partially degraded but was not totally converted into TPA. The hydroxyl group present in the TPA derived from neutral hydrolysis was not observed in the partially degraded PET, as shown in Figure 6. However, some of the peaks at the 1500–1550 and 1320–1380 cm^{-1} wavebands can be attributed to the vibration of the benzene ring in PET and TPA [49]. A strong absorbance peak was noticed at 823 and 742 cm^{-1} , which can be attributed to the para-substituted aromatic ring of TPA and the para-substituted arene group of TPA [28]. The absorbance peaks at 506 and 447 cm^{-1} may correspond to the 1,4-substituted benzene rings [32].

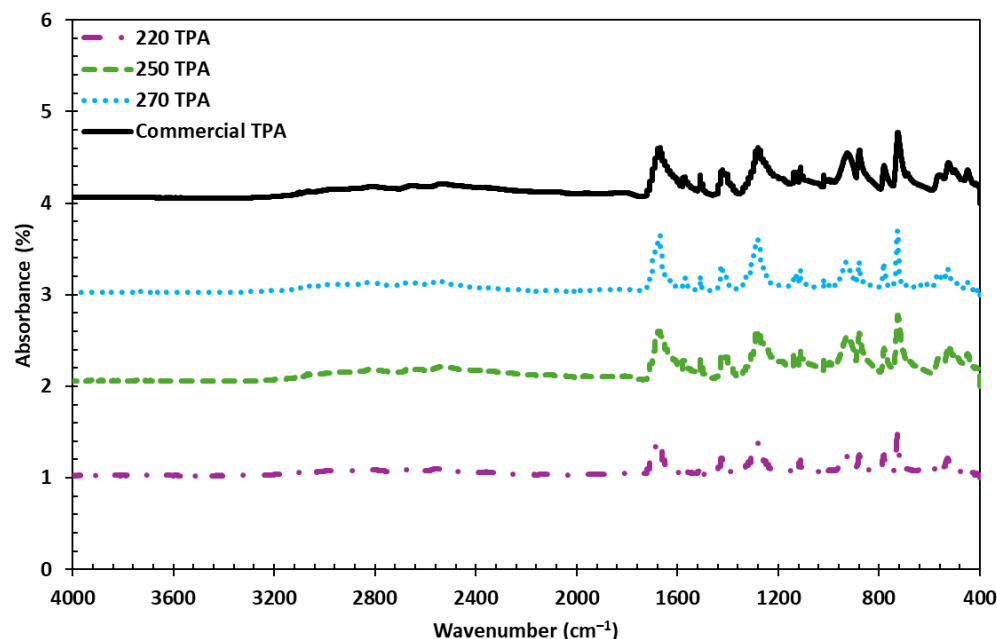


Figure 5. FTIR analysis of TPA derived from PET with a constant residence time of 90 min and PET-to-water ratio of 10.

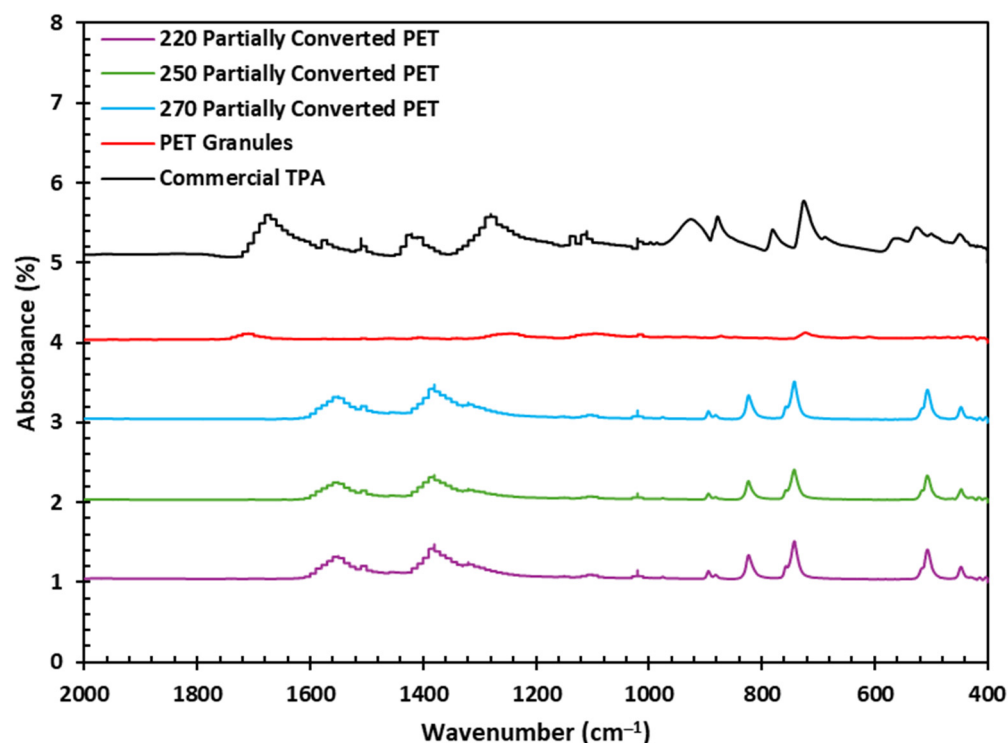


Figure 6. FTIR analysis of PCP with a constant residence time of 90 min and PET-to-water ratio of 10.

In the solid phase extracted after the alkaline treatment, some of the peaks were similar to the commercial TPA and some of the peaks, including those in the hydroxyl group, did not exist. Therefore, it can be concluded that these materials have been partially converted into TPA and are represented as partially converted PET (PCP).

3.5. NMR Analysis of TPA Derived from Neutral Hydrolysis of PET

The TPA obtained from the hydrolysis of PET was subjected to a ^1H -NMR to determine their purity. The results from the NMR analysis indicated that the purity of the TPA derived from the neutral hydrolysis of PET under the studied process conditions ranged between 95.76% and 96.83% in comparison with the commercial TPA. Similar results have been reported in the literature, with TPA purity reaching 97–98% [26,50] and 95.70–98.70% [11]. The variation in product purity could be attributed to the availability of the 2–4% additives, including isophthalic acid, flame-resistant materials, and catalysts, which are added during the polymerisation of TPA and EG to form PET [31].

3.6. Secondary Products Derived from Neutral Hydrolysis of PET

3.6.1. Bis-Hydroxyethyl Terephthalate (BHET)

The esterification of terephthalic acid and ethylene glycol are known to be the building blocks of PET, and bis-hydroxyethyl terephthalate (BHET) may be formed as an intermediate product. It is worth noting that PET may be formed if an excess amount of ethylene glycol is added to BHET [51]. With operating conditions of increased temperature, from 220 °C to 270 °C, at a constant residence time of 90 min, the BHET fraction declined from 25.40 to 7.30%, as shown in Figure 7. One plausible reason for this is that as the residence time increased, the BHET substance decomposed into TPA and EG [52]. Similar results were observed when the temperature was increased to 250 °C.

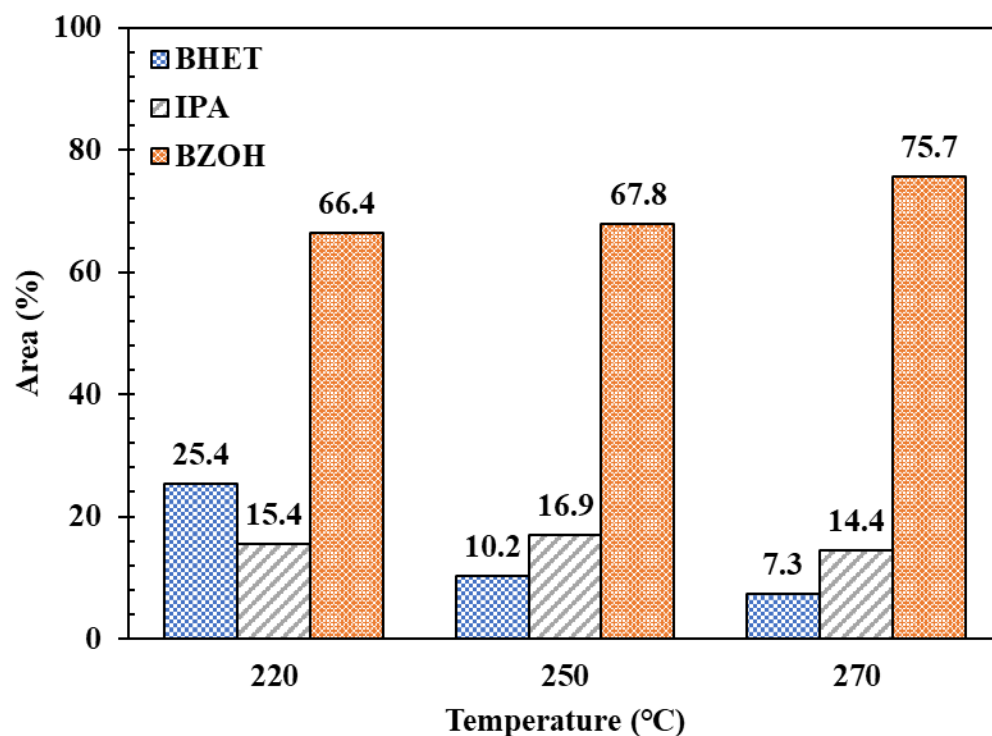


Figure 7. GCMS analysis of aromatic carboxylic acid with a constant residence time of 90 min.

On the contrary, if a reaction temperature beyond the melting point and a residence time between 90 and 105 min were chosen, there was an increase in the BHET fraction from 1.80 to 7.30%. A plausible reason for this could be that the generated TPA and EG in the hydrothermal reactor may have reacted with each other to form BHET [50]. BHET is a valuable product which can be used for the production of PET by adding an excess amount of ethylene glycol [53], and can also be used as an organic precursor to synthesise metal organic frameworks (MOFs) [54].

3.6.2. Isophthalic Acid (IPA)

Isophthalic acid is an isomer of TPA used in the production of polymers, the synthesis of resins, and fibre packaging. Both TPA and IPA are stable aromatic carboxylic acids with low polarity at ambient temperatures [55]. With an increase in temperature, the solubility of TPA and IPA increases. It is worth noting that the degradation pathways for TPA and IPA are similar. When the temperature was increased from 220 °C to 270 °C, there was a decrease in their concentrations, from 15.4% to 14.4%, at a constant residence time of 90 min. This could be due to the molecular rearrangement of TPA (1,4-benzene dicarboxylic acid) into IPA (1,3-benzene dicarboxylic acid) [56]. Another reason could be that the formed IPA could have decarboxylated to form benzoic acid. Isophthalic acid can be used in the synthesis of resins as well as in the production of polymers as a flame retardant [57].

3.6.3. Benzoic Acid (BZOH)

As the temperature was increased from 220 to 270 °C, there was a sharp increase in the BZOH concentration from 66.4 to 75.7%, which could be primarily attributed to the decarboxylation of the TPA. It is worth noting that at higher temperatures, the unstable compound TPA is converted into a stable compound, which is benzoic acid [27,58]. Benzoic acid has been used as a food additive that may improve the gut health of humans when consumed at lower dosages [59].

3.6.4. Phenol

As the temperature was increased from 220 to 270 °C, there was an increase in the formation of phenol from 0.1 to 0.3%. The decarboxylative oxidation of benzoic acid is known to produce either benzene or phenol [60,61]. Further phenol decarboxylation leads to catechol or quinones [62,63].

Phenol is considered as one of the major contaminants that lead to water pollution. Approaches, such as adsorption, biodegradation, chemical oxidation, distillation, enzymatic treatment, and membrane technology, can be employed to remove phenol from the aqueous phase [64].

3.6.5. Ethylene Glycol and Its Derivatives

The secondary products from the degradation of ethylene glycol include ethanol [65], ethane [66], and acetone [67], and were available in trace quantities, as observed in Figure 8. The possible degradation mechanism of ethylene glycol into ethanol, ethane, and acetone is not clearly understood. Further experimentation is necessary to find out the possible degradation mechanism. The traditional pathway by which the degradation of ethylene glycol happens is represented in Figure 9.

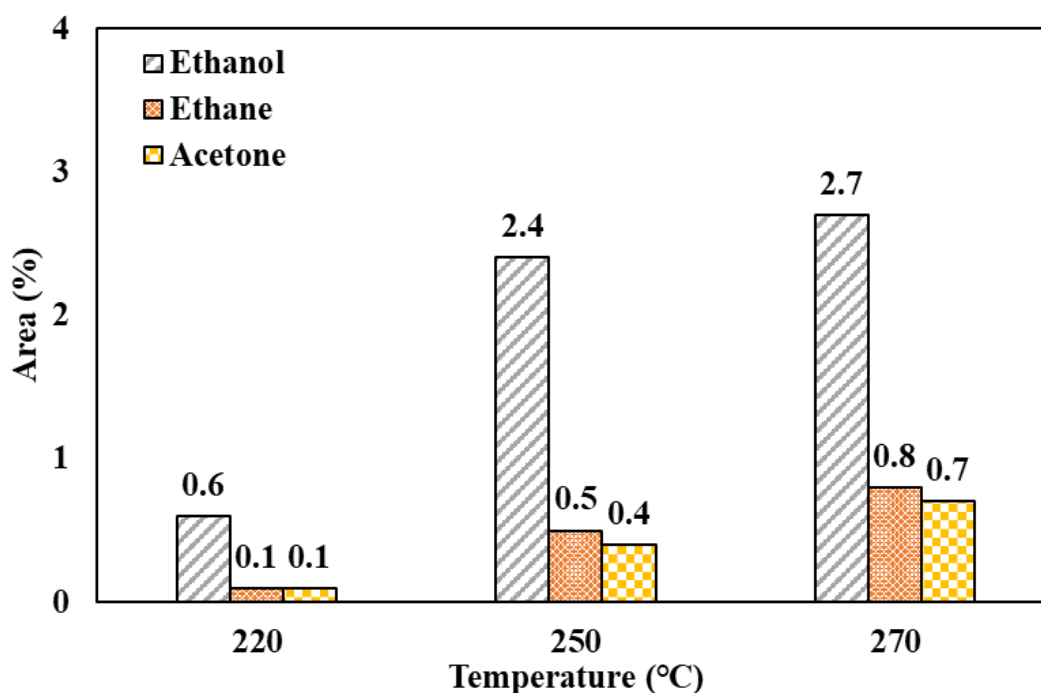


Figure 8. GCMS analysis of EG and its derivatives with a constant residence time of 90 min.

3.7. Insights into the Degradation Pathway of PET into Its Secondary Products Considering the Effect of Temperature and Residence Time

Major Secondary Products

Based on the results, the following degradation routes are discussed. PET is degraded into BHET; then, the further degradation of BHET generates TPA (terephthalic acid). An increased temperature has a favourable effect on the hydrolysis of BHET to form TPA [68,69]. A further increase in temperature leads to its molecular rearrangement to form IPA [70,71], followed by the decarboxylation of the TPA/IPA, leading to the formation of benzoic acid, as shown in Figure 9. Lastly, the benzoic acid might be decarboxylated to form either benzene or phenol [72,73]. The main dominant mechanism during PET degradation is the decarboxylation reaction.

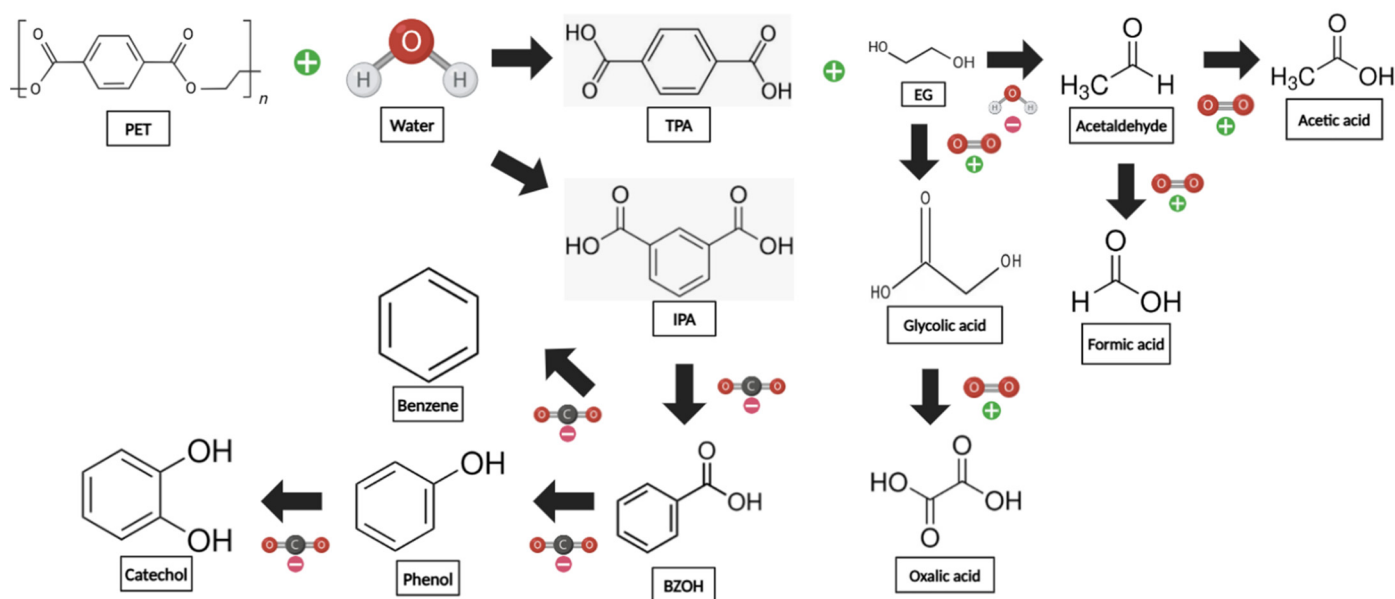


Figure 9. Proposed degradation mechanism of TPA, EG, and their derivatives.

3.8. Green Chemistry Metrics

According to Sheldon et al., the establishment of green chemistry metrics should adhere to the following points:

- (1) The effective use of feedstocks;
- (2) The efficient utilisation of energy during manufacturing;
- (3) The valorisation of waste;
- (4) The elimination of the utilisation of hazardous, toxic, or corrosive solvents;
- (5) The minimal usage of reagents in the synthesis of chemicals [35].

With a temperature and residence time of 250 °C and 90 min, respectively, and an autogenous pressure of 40 bar, the EEI for the PET neutral hydrolysis was estimated to be 5.29×10^4 °C min, which is lower than the literature-reported value of 5.60×10^4 °C min for the neutral hydrolysis of PET using TPA as the catalyst [74], and 6.10×10^4 °C min for neutral hydrolysis with PTSA as the catalyst [29]. When choosing a catalyst, it is important to consider whether the catalyst has the potential to be separated from the product, its recyclable nature, and if it has a lower solubility when used alongside with the solvent [74]. Some of the results reported in the literature are 4×10^4 °C min lower than the estimated EEI (5.29×10^4 – 1.29×10^6 °C min) in the current study, since they used zinc iodide as the catalyst, as shown in Table 1. It is noteworthy that iodine falls under the category of ions, and polar substances could have acted as active intermediate compounds, leading to the complete degradation of the PET and resulting in a higher TPA yield and lower EEI. On the other hand, the value of the estimated EF is in the range of 0.017–0.106, which is comparable to that of the bulk chemicals (EF < 1) [75]. Moreover, scenarios with a lower temperature (220 °C) and residence time (60–90 min) could have higher EEI values between 2.08×10^5 and 1.29×10^6 °C min. The increase in EEI could be due to the lower TPA yield. The order of magnitude of the EEIs for the selected temperatures was identical to the results reported in the literature; for instance, during the neutral hydrolysis of PET, the EEI values were noted as 4.09×10^5 (270 °C, 120 min) and 15.02×10^6 (200 °C, 30 min) [39].

Table 1. Green chemistry metrics of hydrolysis of PET in comparison with the current study.

Temperature	Time	Catalyst	EF	EEC	EEI	References
°C	min			°C ⁻¹ ·min ⁻¹	°C·min	
200	180	TPA	1.486	2.65×10^{-5}	5.60×10^4	[29]
200	120	4-FBA	2.465	2.42×10^{-5}	1.02×10^5	[30]
200	120	ZnI ₂	1.433	3.58×10^{-5}	4×10^4	
200	30	None	50.00	3.33×10^{-6}	15.02×10^6	[39]
270	120		1.124	2.75×10^{-5}	4.09×10^5	
250	30		7.142	1.87×10^{-5}	3.82×10^5	
150	90	PTSA	4.347	7.13×10^{-5}	6.10×10^4	[74]
250	90	None	0.017	3.13×10^{-3}	5.29×10^4	Current study
220	60		0.106	8.23×10^{-4}	1.29×10^6	
220	90		0.034	1.68×10^{-3}	2.08×10^5	
250	30		0.074	2.08×10^{-3}	3.57×10^5	
250	60		0.029	2.62×10^{-3}	1.12×10^5	
270	30		0.038	3.83×10^{-3}	9.73×10^4	
270	60		0.027	2.62×10^{-3}	1.04×10^5	
270	90		0.017	2.34×10^{-3}	7.44×10^4	
300	30		0.046	2.77×10^{-3}	1.67×10^5	

In addition, at 250 °C and residence time of 30–60 min, the EEI values were evaluated to be in the range of 1.12×10^5 to 3.57×10^5 °C min. The lower bound of the values mentioned above is equivalent to the result of 1.02×10^5 °C min where 4-FBA was used as the catalyst [29], and the upper bound matches the result for neutral hydrolysis of 3.82×10^5 °C min [39]. In terms of the type of acid to be utilised as the catalyst, an aromatic carboxylic acid tends to have a positive impact in comparison with an aliphatic carboxylic acid [29].

Overall, it can be found from the previous research that when selecting a catalyst it is important that it falls under the aromatic carboxylic acid group because they have minimal solubility with water and have the potential to be reused again as a catalyst in subsequent cycles [29,30,74]. The higher values of the EEI are due to the lower residence time, between 30 and 60 min, which may be insufficient for the complete degradation of PET during neutral hydrolysis. Similar results were observed at 270 °C for 30–60 min. Lastly, the results at 300 °C for 30 min were reported to be high due to operating under high temperatures above the melting point of PET. Therefore, a lower temperature is necessary for a lower EEI and higher TPA yield [76]. Overall, it can be concluded that the results of the EEI were found to be within the threshold range between 10^4 and 10^6 °C min.

4. Limitations and Recommendations

The PET bottles used in the experiments were used in their pure form. This is not possible in real-world, where contaminants will always be present in PET bottles. Apart from this, the water used for the neutral hydrolysis was deionised MilliQ water, but if it was replaced by tap water or sea water different results could be obtained because of the presence of minerals and metals inherent in non-deionised water. Solar energy-assisted hydrolysis could be an alternative route that could be employed to minimise the energy

required for the neutral hydrolysis process. It is also vital to focus on the impact of changes in thickness, additives, crystallinity, and colouring agents that generally exist in PET bottles. Future investigations related to a techno-economic analysis, with a specific focus on the internal rate of return, payback period, net present value, and sensitivity analysis, will be required to make the current approach scalable.

5. Conclusions

In this current study, PET depolymerisation, via neutral hydrolysis using deionised water, converted a long polyester chain into a solid phase containing TPA and an aqueous phase containing aromatic carboxylic acids. The optimised temperature, time, and pressure for the neutral hydrolysis of PET bottles were found to be 250 °C, 90 min, and 40 bar for a maximum TPA yield of 81.05%. The XPS, FTIR, and NMR analyses of the TPA derived from the neutral hydrolysis process jointly supported that the carbon–oxygen atomic chemistry, surface functional groups, and purity compared well with those of commercial TPA samples.

In addition, the green chemistry metrics that focused on the EF were comparable with those of bulk chemicals (<1–5). The EEI results were analogous to the isothermal hydrolysis of PET, which confirms the efficacy of the current approach. The degradation mechanisms of the aromatic carboxylic acids (BHET, BZOH, and IPA) and ethylene glycol derivatives (ethanol, ethane, and acetone) were proposed. The current approach is environmentally friendly, and it can be used to synthesise TPA with yields of more than 80% and purity in the 95–97% range.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/environments12040127/s1>. Data related to the GCMS analysis was included in the supplementary material S1.

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Abbreviations

The abbreviations used in this manuscript are listed as follows:

BHET	Bishydroxyethyl Terephthalate
BZOH	Benzoic Acid
¹³ C	Stable Isotope of Carbon
EEC	Economy Energy Coefficient
EEI	Environmental Energy Impact

EF	Environmental Factor
EG	Ethylene Glycol
HTT	Hydrothermal Treatment
FBA	Formyl Benzoic Acid
FTIR	Fourier Transform Infrared Spectroscopy
GCMS	Gas Chromatography and Mass Spectroscopy
I.D.	Inner Diameter
IPA	Isophthalic Acid
L	Length
NMR	Nuclear Magnetic Resonance Spectroscopy
PCP	Partially Converted PET
PET	Polyethylene Terephthalate
PTSA	P-Toluene Sulphonic Acid
SDG-12	Sustainable Development Goal 12—Responsible Consumption and Production
TH	Thickness
TPA	Terephthalic acid
XPS	X-ray Photon Spectroscopy
ZnI ₂	Zinc Iodide

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